Taiki Aiba

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Education

Georgia Institute of Technology May 2027

B.S. & M.S. in Computer Science, B.S. & M.S. in Mathematics, B.S. in Applied Languages, Minor in Economics *GPA: 3.96/4.0*

- **Coursework**: OS, Computer Architecture, Processors, Systems/Networks, PLT, Machine/Deep Learning, Statistics
- **Teaching Assistant**: Algorithms (4x), Discrete Mathematics (3x), Multivariable Calculus (1x), Intro to CS I (1x)

Experience

Williams College June 2025 – August 2025

Graduate Research Mentor (NSF REU) *Williamstown, MA*

- Mentored 20 undergraduates on a problem of the game *Quads*, focused on algorithmic optimization and combinatorics.
- Implemented recursive and optimized C++ scripts, utilizing the Moser-de Bruijn sequence to explore its significance.
- Optimized script with Rust via multithreading and regular exit condition checks, achieving a 4x+ performance speedup.
- Delivered mathematical proofs and exact bounds on the result for decks of size a power of 2 at the 2025 and 2026 JMM.

Amazon June 2024 – August 2024

Software Development Engineer Intern *Boston, MA*

- Built a DES in SimPy to model behavior in Alexa's nodes under varying packet loads, with experiment reproducibility.
- Engineered C++ automation tooling to interface with internals, feeding packets of variable sizes to measure latency.
- Processed and statistically analyzed 100k+ CSV/JSON logs with Python to capture patterns and generate insights.
- Fitted empirical models (85% accuracy) to latency data with Matplotlib, informing scalability to large-scale workloads.

Georgia Tech School of Electrical and Computer Engineering August 2023 – December 2023

Research Assistant *Atlanta, GA*

- Translated docs of the PC-FX game console, storing 12 files of 15k+ lines to a GitHub repository for general usability.
- Researched the PC-FX's address map, register list, and I/O access space and the specs of its sound processing device.
- Analyzed the C Compiler of the GMAKER Starter Kit (processing flow/registers), allowing user-made software to run.
- Wrote graphical software in C with the kit to run on the PC-FX to showcase its usability to a team of 3 and a professor.

Projects

De Mathematics Competitions (Founder and Director) | *Jekyll, JavaScript, CSS, Python*

- Designed a GitHub Pages platform using Jekyll, JavaScript, and CSS that hosts math contests, solutions, and videos.
- Crafted 300+ problems for 10+ live contests taken by 500+ students worldwide, earning an average quality of 4.8/5.
- Wrote a Python automation script to generate contests with 80 problems, reducing creation time by 98% (hrs to mins).
- Maintained version control, releases, and documentation through a GitHub repository, spanning throughout 5+ years.

Court Scheduling App (4th Place at GT CS Junior Design Expo) | *TypeScript, Firebase, Expo, Node.js*

- Led backend work in a team of 5 for a court reservation app deployed with Expo (React Native) for 1000+ residents.
- Integrated waitlist class/fields in Firestore with FCFS scheduling/cancellation logic in TypeScript for seamless testing.
- Utilized EmailJS to send reservation emails and Firebase Auth. for verification, offering a robust notification system.

Vibe Doctor | *JavaScript, MongoDB, React, Node.js*

- Deployed a site in a team of 6, utilizing HuggingFace/Spotify APIs, to detect moods to recommend Spotify playlists.
- Leveraged MongoDB Atlas to create an account management system for secure and persistent storage of user profiles.
- Developed 20+ Jest unit tests, ensuring rigorous edge case and exception handling of UI components and login system.

Ceff: A Small C-like Language with Effects | *Python, C*

- Co-invented Ceff, a C-like language with a built-in effect system to combine paradigms (error handling, iterators, etc.).
- Developed a compiler with Python that makes a Ceff program C-compliant through an AST transformation pipeline.
- Reduced latency on 5000 socket connections from 3352 ms on a synchronous C server to 55 ms on a Ceff server.

Graph Neural Networks for Recommender Systems | *PyTorch Geometric, pandas, Matplotlib*

- Implemented a GraphSAGE-based graph autoencoder recommender in PyTorch Geometric on MovieLens 100K dataset.
- Leveraged edge features, using BERT to embed users/movies and computing similarity scores with cosine similarity.
- Achieved an RMSE of 1.0179 (vs baseline 1.0644), nearing the state-of-the-art models UBCF (0.978) and IGMC (0.905).

Skills

Languages: C/C++, Python 3, Java, Go, Rust, JavaScript/TypeScript, SQL, R, Verilog/SystemVerilog
Machine Learning: NumPy, pandas, PyTorch, TensorFlow, scikit-learn, SimPy, Matplotlib
Technologies: React, Expo, Node.js, Spring Boot, Firebase, Docker, GCP, MongoDB, Git, Bash, Linux, WSL, Jira
Certifications: Akuna Capital – Options 201, MIT Professional Learning – No Code AI and Machine Learning
Awards: USACO Silver, Codeforces Specialist (top 8%), 24 on Putnam (606th/4000+), 10 on AIME twice (top 4%)